

Installing Plotly

Install Plotly using pip, just as you did for Matplotlib:

```
$ python -m pip install --user plotly
$ python -m pip install --user pandas
```

Plotly Express depends on *pandas*, which is a library for working efficiently with data, so we need to install that as well. If you used **python3** or something else when installing Matplotlib, make sure you use the same command here.

To see what kind of visualizations are possible with Plotly, visit the gallery of chart types at <https://plotly.com/python>. Each example includes source code, so you can see how Plotly generates the visualizations.

Creating the Die Class

We'll create the following Die class to simulate the roll of one die:

```
die.py  from random import randint

class Die:
    """A class representing a single die."""

    ❶ def __init__(self, num_sides=6):
        """Assume a six-sided die."""
        self.num_sides = num_sides

    def roll(self):
        """Return a random value between 1 and number of sides."""
    ❷    return randint(1, self.num_sides)
```

The `__init__()` method takes one optional argument ❶. With the Die class, when an instance of our die is created, the number of sides will be six if no argument is included. If an argument *is* included, that value will set the number of sides on the die. (Dice are named for their number of sides: a six-sided die is a D6, an eight-sided die is a D8, and so on.)

The `roll()` method uses the `randint()` function to return a random number between 1 and the number of sides ❷. This function can return the starting value (1), the ending value (`num_sides`), or any integer between the two.

Rolling the Die

Before creating a visualization based on the Die class, let's roll a D6, print the results, and check that the results look reasonable:

```
die_visual.py  from die import Die

# Create a D6.
❶ die = Die()
```
